

Population viability analysis and sensitivity analysis



Population Viability Analysis

The use of quantitative methods to predict the likely future status of a population or collection of populations of conservation concern.

(Morris and Doak 2002)

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Now routine part of species assessments and recovery plans

USES OF PVA (FROM MORRIS AND DOAK (2002))

- (1) Assessing the extinction risk of a single population
- (2) Comparing relative risks of two or more populations
- (3) Analyzing and synthesizing monitoring data
- (4) Identifying key life stages or demographic processes as management targets (sensitivity analysis)
- (5) Determining how large a reserve needs to be to achieve a desired level of protection from extinction
- (6) Determining how many individuals to release to establish a new population
- (7) Setting limits on the harvest or “take” from a population that are compatible with its continued existence
- (8) Determining how many (and which) populations are needed to achieve a desired overall likelihood of species persistence

- (1) Develop objectives
- (2) Develop a set of competing models
- (3) Design a study to collect necessary data
- (4) Fit models to data and select the best model(s)
- (5) Use model(s) to identify best management options
- (6) Implement management option, monitor the consequences, and refine models
- (7) Return to step (4)

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Very similar to adaptive management

(1) Count-based PVA

- ▶ All you have is estimates of abundance in each year
- ▶ This is the cheapest, but least-informative method

(2) Demographic PVA

- ▶ Requires estimates of vital rates
- ▶ Useful for identifying key demographic parameters

(3) Metapopulation viability analysis

- ▶ Useful in reserve design

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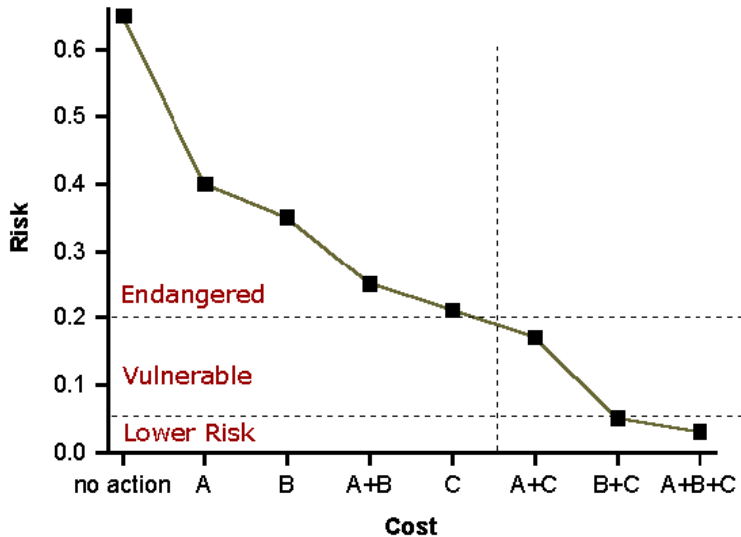
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All methods require clear definition of time horizon and acceptable level of extinction risk

PRODUCTS OF PVA

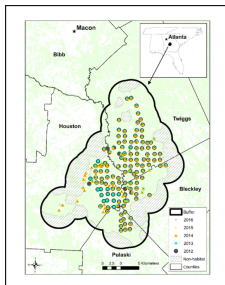


CENTRAL GEORGIA BLACK BEARS

Smallest and most isolated bear population in GA.

Little information about the effects of harvest on population viability.

Five years of genetic capture-recapture data were used to estimate abundance and develop population models.



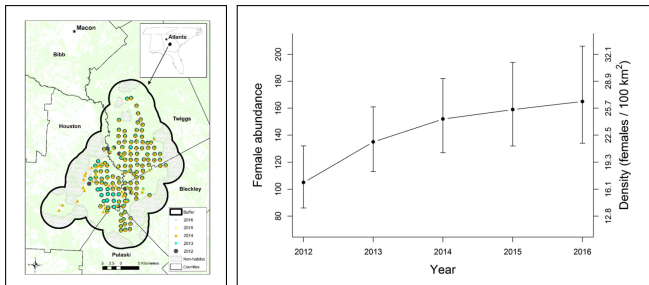
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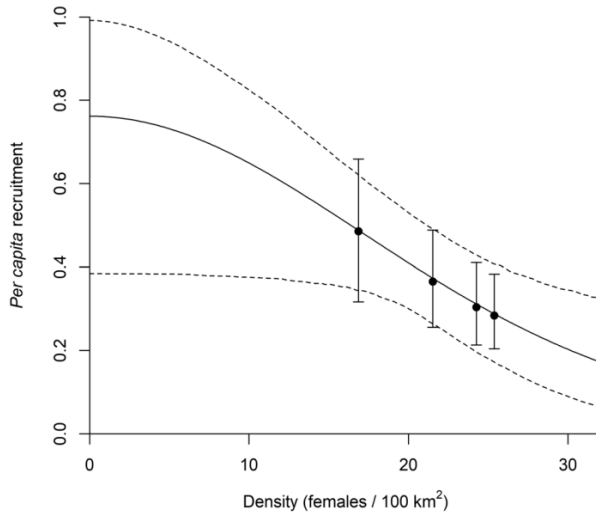
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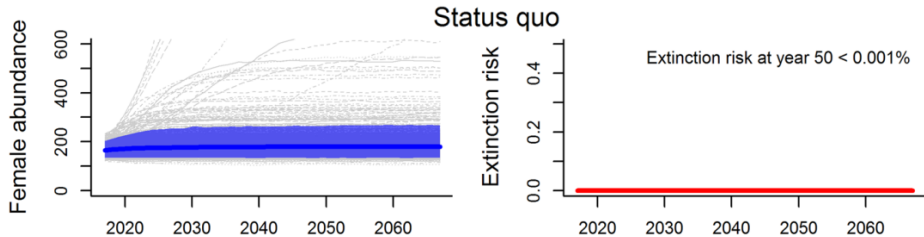


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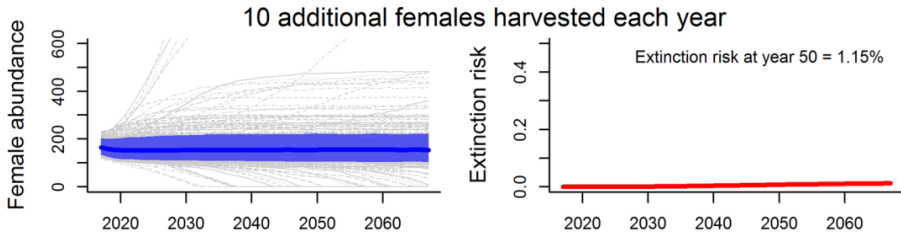
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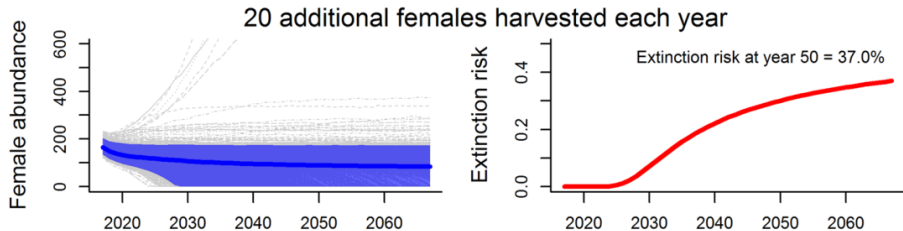
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Sensitivities allow us to make statements such as:

“Increasing subadult fecundity by a small amount increases λ by 0.01, whereas increasing adult fecundity by the same amount increases λ by 0.02. Therefore, population growth is more sensitive to changes in adult fecundity.”

SENSITIVITY EXAMPLE

Parameter	$\Delta\theta$	$\Delta\lambda$	Sensitivity
Fecundity of first age class (f_1)	0.05	0.010	0.20
Fecundity of second age class (f_2)	0.05	0.003	0.06
Survival of first age class (s_1)	0.05	0.040	0.80
Survival of second age class (s_2)	0.05	0.030	0.60

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Examples

“A 1% increase in f_1 increases λ by 0.01%.”

“A 1% increase in s_1 increases λ by 0.05%.”

Suppose we have the following stage-structured projection matrix:

```
A <- matrix(c(
  0.2, 0.8, 1.0, 0.9,
  0.4, 0.0, 0.0, 0.0,
  0.0, 0.6, 0.0, 0.0,
  0.0, 0.0, 0.8, 0.5), nrow=4, byrow=TRUE)
```

We can use eigenanalysis to compute λ , stable age distribution, reproductive value, sensitivities, and elasticities.

```
eA <- eigen(A)
lam <- Re(eA$values[1])
lam                                     ## Asymtotic growth rate

## [1] 1.034908

w <- Re(eA$vectors[,1])
w <- w/sum(w)                          ## Stable age distribution

v <- Re(eigen(t(A))$vectors[,1])      ## Reproductive value
```

The sensitivities are given by $z_{ij} = \frac{v_i w_j}{\mathbf{v} \mathbf{w}}$

```
z <- outer(v,w)/c(v%*%w)
```

```
z ## Sensitivites
```

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 0.3473925 0.1342698 0.07784447 0.1164229
## [2,] 0.7251023 0.2802575 0.16248251 0.2430061
## [3,] 0.7875009 0.3043751 0.17646493 0.2639179
## [4,] 0.5844985 0.2259131 0.13097572 0.1958851
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```

The elasticities are given by $e_{ij} = \frac{a_{ij} z_{ij}}{\lambda}$

```
e <- A*z/lam
e ## Elasticities

##           [,1]      [,2]      [,3]      [,4]
## [1,] 0.06713492 0.1037926 0.0752187 0.10124623
## [2,] 0.28025755 0.0000000 0.0000000 0.00000000
## [3,] 0.00000000 0.1764649 0.0000000 0.00000000
## [4,] 0.00000000 0.0000000 0.1012462 0.09463883
```

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Sensitivity analysis is a component of PVA that can guide conservation planning.

- Sensitivity measures the change in λ given an absolute change in a parameter.
- Elasticity measures the proportional change in λ given a proportional change in a parameter.

Problems with PVA

Most PVAs are “*essentially games played with guesses.*” (Caughley, G. 1994. Conservation Biology).

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Solutions

Collect better data! We need more demographic research, conducted in an adaptive management framework.

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Solutions

Collect better data! We need more demographic research, conducted in an adaptive management framework.

When data are limited (and they always are), decrease the time horizon and acknowledge uncertainty in decision process.